Specification

Technical Field

The present invention belongs to the technical fields of artificial intelligence and network security, and specifically relates to a three-layer separated multi-agent system operation system and method based on **trusted isolated operating environments**, time-sequence replay prevention and memory **physical degaussing erasure**.

Background Art

With the boom of Large Language Models (LLMs) and Multi-Agent Systems (MAS), multi-agent collaboration has been widely applied to complex business scheduling. However, under existing architectures, when an Agent executes tasks, it is usually required to decrypt the user's private knowledge base (such as confidential enterprise data and personal privacy documents) directly and inject the plaintext data into the cloud-based LLM inference engine. This mode leads to severe security hazards:

1. Loss of user data sovereignty: The user's core confidential data is retained in plaintext or semi-plaintext form on the computing power service provider's (e.g., LLM cloud platform) side. Once the computing node is compromised or subjected to silent malicious auditing, data leakage occurs. In traditional local vector library Retrieval-Augmented Generation (RAG) solutions, data is deeply coupled with LLM inference logic, which physically enables computing power providers to access the plaintext context of the user's knowledge base.

2. Cold Boot Attack via side-channel charge residue: When physical hosts of LLM engines process large batches of context, sensitive data resides in Dynamic Random-Access Memory (DRAM). Existing Trusted Execution Environment (TEE) solutions only implement logical/ciphertext isolation for CPUs or virtual memory at the software layer, without performing charge residue erasure on physical DRAM when the security boundary is breached or tasks are terminated. Attackers can recover complete private knowledge data by reading residual charges in DRAM within a short window after power-off via physical freezing or specific voltage side-channel attacks.

3. **Over-ingestion Attack** (the technical implementation of Semantic Binge at the system level): Malicious agents can construct infinitely nested prompt words or malicious instruction streams to trick the execution engine into continuously consuming context cache, thereby crippling system performance and conducting unauthorized side-channel extraction of data.

Therefore, it is an urgent technical problem to design a secure multi-agent operation system that completely decouples data reading/writing, rule judgment and general LLM computing power at both physical and network levels, and is equipped with **physical degaussing erasure** and network fingerprint forensics capabilities.

Summary of the Invention

Technical Problems to Be Solved

The present invention aims to solve the technical problems of unauthorized disclosure vulnerabilities of users' private knowledge bases on cloud computing nodes, risks of cold boot side-channel extraction of DRAM charge residues, and lack of active fingerprint fusing forensics means for system gateways when facing **Over-ingestion Attacks** in existing multi-agent collaborative architectures.

Technical Solutions

To solve the above technical problems, the present invention provides a three-layer separated multi-agent system operation method.

Aspect 1: Operation Method Independently Executed by the Rule Cognition Layer

The **Rule Cognition Layer** is physically deployed in a **trusted isolated operating environment**, and the method includes the following steps:

1. Receive external agent interaction requests, and apply for allocation of a temporary **isolated secure operating domain** in the **trusted isolated operating environment**;

2. Start a hardware lifetime clock timer bound to the **isolated secure operating domain**, where the initial value of the hardware lifetime clock timer is a preset lifetime threshold **Timer_TTL**;

3. Send a request to the **Isolated Asset Layer** via a **unidirectional read-only physical tunnel** to read private knowledge slice data corresponding to the interaction request;

4. Within the **isolated secure operating domain**, aggregate and calculate the read private knowledge slice data with a preset standardized cognitive protocol instruction set to generate a target control instruction stream;

5. Invoke a secure hash algorithm to perform asymmetric hash coupling calculation on the microsecond execution timestamp **T_μs** of the target control instruction stream, a logical entropy value **H_logic** and a session symmetric key **K_session**, so as to generate an **instruction anti-replay digital fingerprint** serving as the causal birth certificate of instructions;

6. Unidirectionally output the target control instruction stream and its corresponding **instruction anti-replay digital fingerprint** to the **Computing Execution Layer** for secure inference calculation by the **Computing Execution Layer**;
7. When the hardware lifetime clock timer expires, or penetration of the boundary of the **trusted isolated operating environment** is detected, a hardware security microcontroller forcibly triggers a **physical degaussing erasure** interrupt to perform overwrite erasure on physical memory of the operating domain loaded with private knowledge slice data.

Preferably, the **trusted isolated operating environment** includes at least one of a hardware-level Trusted Execution Environment (TEE), a hardware virtual machine-level confidential computing enclave, a virtualized secure container, and a kernel-level logical secure domain.

Preferably, the **isolated secure operating domain** includes at least one of an enclave sandbox based on hardware memory encryption, an isolated enclave based on lightweight hardware virtualization, and a logical isolated secure zone based on a microkernel.

Preferably, the step of performing **physical degaussing erasure** overwrite on physical memory of the operating domain loaded with private knowledge slice data includes:
1. Execute no less than three rounds (**N_scrub** ≥ 3) of physical overwrite cycles: write all physical addresses corresponding to the operating domain physical memory with all-zero value **0x00** in the first round; write all-one value **0xFF** in the second round; write random noise data generated by a cryptographically secure pseudo-random number generator in the third round;
2. Invoke the processor cache line flush instruction _mm_clflush to forcibly flush data in the processor cache back to Dynamic Random-Access Memory (DRAM) and complete charge neutralization, so as to block cold boot side-channel extraction of charge residues.

Preferably, the generation formula of the **instruction anti-replay digital fingerprint** is as follows:

```
OutputHash = HMAC-SHA256(T_μs || H_logic, K_session)
```

Where HMAC-SHA256 represents a keyed-hash message authentication code function based on the SHA256 hash algorithm; `||` denotes a byte-level concatenation operation of the microsecond timestamp **T_μs** and the logical entropy value **H_logic**.

Preferably, the calculation formula of the logical entropy value **H_logic** is as follows:

```
H_logic = - ∑_i=1^M P(o_i) log₂ P(o_i)
```

Where P(o_i) represents the occurrence probability distribution of a single control instruction opcode o_i in the target control instruction stream extracted within the **isolated secure operating domain**; M is the size of the deduplicated instruction set of extracted control opcodes.

Preferably, the **unidirectional read-only physical tunnel** adopts the AES-256-GCM symmetric encryption algorithm to perform session-level dynamic key symmetric encryption on transmission data streams in the read-only physical tunnel.

Preferably, the physical upper limit of the lifetime threshold **Timer_TTL** is set to 120 seconds.

Preferably, the hardware lifetime clock timer includes at least one of a hardware clock timer, a secure virtual clock deployed within the **trusted isolated operating environment**, a logical software timer, and a high-precision secure clock source.

Aspect 2: Operation Method Independently Executed by the Computing Execution Layer

The **Computing Execution Layer** is physically deployed on a computing engine server, and the method includes the following steps:
1. Receive the target control instruction stream and its corresponding **instruction anti-replay digital fingerprint** sent by the **Rule Cognition Layer**;
2. Parse the **instruction anti-replay digital fingerprint**, extract the execution timestamp therein and compare it with the current system time; if the time difference exceeds a preset anti-replay delay window, reject inference invocation of the target control instruction stream;
3. After passing anti-replay verification, perform inference calculation on the target control instruction stream via a general large language model engine, and monitor whether the data throughput of a single session exceeds a preset session buffer upper limit **Theta_max**;
4. If the session buffer upper limit **Theta_max** is exceeded, determine that an **Over-ingestion Attack** occurs, immediately fuse the session, and force the underlying communication protocol stack to send a TCP Reset (RST) packet carrying a specific window field value **W_forensic** to broadcast forensic characteristic fingerprints externally.

Preferably, the physical setting of the session buffer upper limit **Theta_max** is 4096 Tokens.

Preferably, in the TCP Reset packet carrying forensic characteristic fingerprints, the Window Size field value **W_forensic** in the TCP header is forcibly locked to a preset fixed hexadecimal constant **oxABCD**.

Aspect 3: Operation Method Independently Executed by the Isolated Asset Layer

The **Isolated Asset Layer** is deployed on a local data storage node under user sovereignty control, and the method includes the following steps:

1. Establish a local Inter-Process Communication (IPC) secure tunnel with the **Rule Cognition Layer**, and bind an access control whitelist within the secure tunnel;

2. Receive a data reading request sent by the **Rule Cognition Layer** via the **unidirectional read-only physical tunnel**, and verify the session digital signature in the request;

3. After signature verification passes, decrypt private knowledge slice data corresponding to the request in local secure zone memory, and map the data to the **unidirectional read-only physical tunnel** in read-only mode for unidirectional reading by the **Rule Cognition Layer**;

4. When closure or disconnection of the **unidirectional read-only physical tunnel** is detected, forcibly reclaim mapped memory pages and perform no less than three rounds of **physical degaussing erasure**.

Aspect 4: Three-Layer Collaborative Multi-Agent System Operation Method

The method is collaboratively executed by the **Isolated Asset Layer**, the **Rule Cognition Layer** and the **Computing Execution Layer**, and includes the following steps:

1. The **Rule Cognition Layer** receives external agent interaction requests, applies for allocation of a temporary **isolated secure operating domain** in the **trusted isolated operating environment**, and starts a lifetime timer bound to the **isolated secure operating domain**;

2. The **Rule Cognition Layer** sends a request to the **Isolated Asset Layer** via a **unidirectional read-only physical tunnel** to read corresponding private knowledge slice data;

3. The **Isolated Asset Layer** decrypts the private knowledge slice data and maps the data to the **unidirectional read-only physical tunnel**;

4. Within the **isolated secure operating domain**, the **Rule Cognition Layer** aggregates and calculates the private knowledge slice data with a preset instruction set to generate a target control instruction stream, generates a corresponding **instruction anti-replay digital fingerprint**, and unidirectionally outputs the fingerprint to the **Computing Execution Layer**;

5. After the **Computing Execution Layer** verifies the **instruction anti-replay digital fingerprint**, a general inference engine performs inference calculation on the target control

instruction stream;
6. When the task is completed, the timer expires, or penetration of the security boundary is detected, the **Rule Cognition Layer** and the **Isolated Asset Layer** respectively perform **physical degaussing erasure** on temporary memory occupied by each layer.

Preferably, the lifetime timer includes at least one of a hardware clock timer, a secure virtual clock deployed within the **trusted isolated operating environment**, a logical software timer, and a high-precision secure clock source.

Aspect 5: Three-Layer Separated Multi-Agent System Operation System

The system includes a physically isolated **Rule Cognition Layer**, an **Isolated Asset Layer** and a **Computing Execution Layer**, where:
- The **Rule Cognition Layer** is physically deployed in a **trusted isolated operating environment**, and is configured to execute any method described in Aspect 1 (i.e., the method according to any one of **Claim 1** to **9**);
- The **Isolated Asset Layer** is physically deployed on a local data storage node, and is configured to execute the method described in Aspect 3 (i.e., the method according to **Claim 13**);
- The **Computing Execution Layer** is physically deployed on a computing engine server, and is configured to execute any method described in Aspect 2 (i.e., the method according to any one of **Claim 10** to **12**).

Aspect 6: Computer-Readable Storage Medium

A computer-readable storage medium stores a computer program thereon, **characterized in that** the program, when executed by a processor, implements any method according to any one of **Claim 1** to **15**.

Advantageous Effects

The beneficial effects brought by the present invention include:
1. Physical-level data sovereignty control via three-layer decoupling: The multi-agent system is split into an **Isolated Asset Layer**, a **Rule Cognition Layer** and a **Computing Execution Layer**, which completely physically decouples "knowledge storage", "instruction assembly and rule control" and "large-scale computing generation". The general large-scale computing execution engine only receives desensitized control instruction streams, and cannot access plaintext private knowledge bases in physical form or throughout the entire lifecycle. This fundamentally solves the

problem of silent snooping and theft of user data privacy by large cloud computing power platforms from the physical topology perspective.

2. Physically eliminate risks of memory charge residue leakage: When the **isolated secure operating domain** is closed or the boundary is penetrated, the microcontroller triggers hardware-level **physical degaussing erasure**. Multiple rounds of physical overwrite cycles are adopted to forcibly rewrite DRAM pages occupied by the operating domain, and the clflush instruction is used to force CPU cache write-back and charge neutralization, which physically blocks cold boot side-channel extraction attacks based on residual charges.

3. Anti-replay and active security fusing forensics: An **instruction anti-replay digital fingerprint** based on adaptive weighted information entropy is generated for time-sequence interception. Sessions exceeding the buffer upper limit caused by **Over-ingestion Attacks** are actively disconnected, and a specific fingerprint tag (**0xABCD**) is injected into TCP Reset packets as forensics evidence. External traffic capture tools can identify infringement without reviewing source code of cloud vendors, which greatly improves the feasibility of security supervision and patent rights protection.

4. Anti-circumvention protection for pure software distribution: By adding claims covering computer-readable storage media, any software packages, SDKs or firmware containing the multi-agent operation instructions constitute infringement subjects when separately distributed, sold or transmitted over networks, which fully enhances infringement rights protection effectiveness in commercial circulation scenarios.

Brief Description of the Drawings

Figure 1 is a physical architecture schematic diagram of a three-layer separated multi-agent system operation system provided by an embodiment of the present invention.

Figure 2 is a flow schematic diagram of a multi-agent system operation method executed by the **Rule Cognition Layer** provided by an embodiment of the present invention.

Figure 3 is a flow schematic diagram of a multi-agent system operation method executed by the **Computing Execution Layer** provided by an embodiment of the present invention.

Figure 4 is a flow schematic diagram of a multi-agent system operation method executed by the **Isolated Asset Layer** provided by an embodiment of the present invention.

Detailed Description of Embodiments

5.0 Alignment Definition of Sovereign Nouns

1. **Rule Cognition Layer**: also referred to as "logical tunnel control core", a secure hub for knowledge assembly and trusted logical shunting, functionally corresponding to a standardized protocol parsing node.
2. **Isolated Asset Layer**: also referred to as "physical sovereign storage zone", physically deployed on local data storage nodes, functionally corresponding to a standard storage protocol publishing node.
3. **Computing Execution Layer**: also referred to as "LLM inference engine", deployed on general cloud computing platforms, functionally corresponding to a standard execution node.
4. Logic Actor: a consistent logical judgment entity carrying cognitive instruction sets within an **isolated secure operating domain**.
5. **Instruction anti-replay digital fingerprint**: also referred to as "instruction physical birth certificate", a signature ensuring unidirectional time sequence and causal self-consistency of control flows, functionally corresponding to a standardized protocol publishing node.

5.1 Physical Topology of the Three-Layer Decoupled System

This three-layer separated physical topology is the engineering implementation of the Three-Layer Architecture defined in Item 117 of the terminology dictionary: Isolated Asset Layer = Human Asset Layer, Rule Cognition Layer = Human Cognitive Layer, Computing Execution Layer = AI Execution Layer.

As shown in **Figure 1**, the operation system in this embodiment adopts a three-layer physically isolated topology. The **Isolated Asset Layer** is physically deployed in a local data storage center under the user's local area network, and stores un-desensitized private texts, vector libraries and sovereign data. The **Rule Cognition Layer** communicates with the **Isolated Asset Layer** via a dedicated sovereign gateway. This layer is deployed on a local security boundary gateway equipped with a **trusted isolated operating environment** (such as Intel SGX/ARM TrustZone TEE chips, virtual machine-level confidential computing enclaves, virtualized secure containers or kernel secure domains). It assembles knowledge into Logic Actor instruction packages and never allows plaintext data to cross its trusted physical boundary. The **Computing Execution Layer** is a remote computing center (such as a cloud AI cluster), which is only responsible for generating

Tokens based on restricted logical instructions output by the Cognition Layer, and does not access original knowledge data throughout the entire lifecycle.

As a deployment embodiment, the **Rule Cognition Layer**, the **Isolated Asset Layer** and the **Computing Execution Layer** of the present invention can be physically deployed on mutually independent physical nodes (for example, the **Isolated Asset Layer** is deployed on a local private sovereign storage terminal, the **Rule Cognition Layer** is deployed on the user's edge gateway or local hardware gateway, and the **Computing Execution Layer** is deployed on a third-party public cloud server); they can also be flexibly distributed across devices or networks. In addition, the three-layer architecture of the present invention can also be deployed on different logical/secure partitions of a single physical device (such as a hardware TEE partition, confidential storage partition and general CPU/GPU inference partition of a single intelligent terminal device, i.e., single-device multi-partition deployment). Physical or logical complete decoupling and separation are realized via process isolation, hardware bus-level partitioning or secure microkernel isolation strategies, covering various deployment scenarios ranging from edge computing and embedded end-side AI to large-scale cloud collaboration.

As an anti-circumvention characteristic design of the architecture, merging any two layers of the three-layer structure (the **Rule Cognition Layer**, the **Isolated Asset Layer**, the **Computing Execution Layer**) in actual deployment (for example, deploying the **Rule Cognition Layer** and the **Computing Execution Layer** on a public cloud computing node, or deploying the **Isolated Asset Layer** and the **Rule Cognition Layer** on an untrusted computing node) will directly lead to the collapse of the following technical defects and system defense lines:
(1) Plaintext private knowledge will be directly exposed to physical memory of the untrusted host where the large-scale inference engine is located, causing the user to completely lose control of data sovereignty;
(2) The physical/process-level decoupled unidirectional channel is destroyed, enabling cold boot side-channel charge residue attacks to directly extract large batches of knowledge data residing in memory, and the **physical degaussing erasure** interrupt (**N_scrub**) becomes invalid due to loss of physical security boundary judgment;
(3) Boundary penetration detection capability fails, and abnormal virtual page fault monitoring of the host cannot effectively isolate external malicious overflow detection.

Such compromised schemes with merged layers are in fact the fatal defects of the prior art solved by the present invention. The present invention enforces physical/logical decoupling of the three

layers to ensure that the full data security boundary remains intact even if a single-layer device is compromised.

5.2 Processing Details of the Rule Cognition Layer

As shown in **Figure 2**, the control steps of the **Rule Cognition Layer** when running multi-agents are as follows:

1. Dynamic application for an **isolated secure operating domain**: When an agent task request is received, the **trusted isolated operating environment** (such as a hardware TEE or virtual machine secure enclave) dynamically allocates isolated memory via hardware primitives to instantiate a temporary **isolated secure operating domain** for a Logic Actor (such as a physical secure sandbox, isolated secure operating container or microkernel secure zone).

2. Lifetime clock restriction: A hardware security chip initializes a decrement-only lifetime timer **Timer_TTL**, whose initial configurable range is 30s to 300s (e.g., 30s, 60s, 90s, 120s, 300s; the default setting in this embodiment is 120 seconds). If the task is not completed within the set time, a hardware-level interrupt is unconditionally triggered to close the isolated domain, preventing malicious instructions from occupying secure resources infinitely for bypass side-channel reading. As a hardware-level degradation or software-level compatible implementation of the trusted clock, in alternative embodiments, the lifetime clock timer can be replaced or supplemented with a secure clock within a trusted isolated environment (such as an enclave), a trusted virtual clock, a logical software timer or a high-precision secure clock source besides a hardware timer.

3. Reading via **unidirectional read-only physical tunnel**: An AES-256-GCM session key negotiated between the local end and the asset storage node is used to establish a **unidirectional read-only physical tunnel**, and only partial private knowledge slice data required for the task is loaded into the memory of the **isolated secure operating domain**.

4. Instruction stream and birth certificate generation: Within the **isolated secure operating domain**, private data and instruction sets are assembled into a control instruction stream. To prevent interception and replay by intermediate network nodes, a secure microcontroller calculates an **instruction anti-replay digital fingerprint** with the algorithm:

```
OutputHash = HMAC-SHA256(T_μs || H_logic, K_session)
```

Where `||` denotes byte-level concatenation of the microsecond timestamp **T_μs** and logical entropy **H_logic**. The logical entropy **H_logic** is adaptively weighted and calculated based on information theory via the following formula:

```
H_logic = - ∑_i=1^M P(o_i) log₂ P(o_i)
```

Where P(o_i) represents the occurrence probability distribution of a single control instruction opcode o_i in the target control instruction stream extracted within the **isolated secure operating domain**; M is the size of the deduplicated instruction set of extracted control opcodes. Due to the addition of the microsecond-level timestamp **T_μs**, the Computing Layer can verify the time sequence within an extremely narrow delay window, and any intercepted historical instruction automatically fails once exceeding the window.

5. **Physical degaussing erasure** and dual hardware-software boundary protection: When the **isolated secure operating domain** exits, a connection abnormality occurs, or penetration of the boundary of the **trusted isolated operating environment** is detected, the system immediately suspends all calculations and initiates physical degaussing and charge neutralization procedures. Penetration of the boundary of the **trusted isolated operating environment** not only guards against hardware-level physical shell disassembly attacks and voltage/clock glitch injection attacks, but also includes software-level abnormal unauthorized page reading detection of enclave memory, speculative execution side-channel vulnerability (such as Meltdown or Spectre overflow detection), and software/firmware-level boundary abnormal penetration such as abnormal disconnection of asymmetric key session heartbeats.

An eBPF monitor is deployed in the kernel mode of the host operating system to capture the frequency of abnormal access page fault interrupts targeting isolated physical memory page tables in real time. When the page fault interrupt frequency exceeds a preset security threshold, boundary penetration is determined, and a hardware security microcontroller forcibly triggers a **physical degaussing erasure** interrupt:

A degaussing controller is activated to perform **N_scrub** rounds of overwriting on all addresses of Dynamic Random-Access Memory (DRAM) allocated to the **isolated secure operating domain** (**N_scrub** can be configured between 3 and 7 rounds, with the default **N_scrub** = 3 in this embodiment): in the first round, **0x00** is physically written to discharge charges of all capacitor units to a low potential; in the second round, **0xFF** is physically written to charge all capacitor units to the highest potential; in the third round, physical noise words generated by a Cryptographically Secure Pseudo-Random Number Generator (CSPRNG) are written to completely eliminate physical metastable residues of the medium. If configured for 5 or 7 rounds, alternating writing of all-ones and pseudo-random noise words is repeated in subsequent steps.

After each round of overwriting, the assembly instruction clflush is executed continuously to clear all levels of processor cache, forcing all plaintext data remaining in the cache to be synchronized back to physical memory pages for charge neutralization, preventing cold boot side-channel extraction.

5.3 Verification and Active Fusing of the Computing Execution Layer

As shown in **Figure 3**, the processing steps of the **Computing Execution Layer** after receiving instruction streams from the Cognition Layer are as follows:

1. Anti-replay fingerprint verification: Parse the received OutputHash. The computing gateway extracts the timestamp and compares it with the current NTP time. If the time difference exceeds 500 milliseconds, the instruction is determined as an intercepted historical replay, and invocation failure is returned directly to block inference.

2. **Over-ingestion Attack** detection: The inference engine processes the instruction stream and counts the cumulative number of generated Tokens in real time. The upper limit of the single-session buffer **Theta_max** can be adaptively configured according to the model context window size of the target computing engine (e.g., any Token gear among 1024, 2048, 4096, 8192, 16384; the default limit in this embodiment is 4096 Tokens).

3. Fingerprint-tagged fusing forensics: When the Token consumption curve of a single inference exceeds **Theta_max**, it is determined that the computing layer is subject to an external malicious cyclic **Over-ingestion Attack**. The communication protocol stack of the gateway immediately executes session fusing, and does not send a standard four-way handshake packet; instead, a TCP Reset (RST) packet is forcibly sent. When constructing the TCP RST packet, a socket control function is invoked to forcibly overwrite and fill the Window Size field in the TCP header with a specific forensic magic fingerprint **W_forensic** (such as the preset fixed hexadecimal constant **oxABCD**). As an anti-circumvention extension, the injection position of the forensic magic fingerprint **W_forensic** is not limited to the TCP Window Size; it can also reuse or forcibly overwrite the Reserved bits field, Urgent Pointer field in the TCP header, or a specific checksum slot of a custom application-layer communication packet. Once external network traffic monitoring (such as traffic capture analysis) detects that a specific field matches the preset feature, physical forensic evidence is formed directly, providing direct technical court evidence for rights protection and infringement auditing.

5.4 Process Isolation of the Isolated Asset Layer

As shown in **Figure 4**, the **Isolated Asset Layer** executes the following interaction steps to ensure data remains local:

1. Whitelist handshake: When a **unidirectional read-only physical tunnel** connection is established, the **Isolated Asset Layer** reads a physical certificate bound to hardware of the Cognition Layer.

2. Dynamic memory page mapping: After a data request arrives and signature verification passes, the operating system of the asset layer loads corresponding data into a local encrypted RAM area, and dynamically maps its physical address range to the **unidirectional read-only physical tunnel** with read-only permissions. The **Rule Cognition Layer** is only allowed unidirectional reading, and the asset layer prohibits write operations from the **Rule Cognition Layer**.

3. Erasure upon connection fusing: When disconnection of the unidirectional IPC tunnel is detected, the operating system of the asset layer releases memory page descriptors of mapped physical memory locally, and performs three rounds of random noise overwriting (configurable between 3 and 7 rounds) on involved physical memory units to block leakage of residual electromagnetic features.

5.5 Vertical Industry Application Scenarios and Embodiments

The three-layer separated operation method and system of the present invention can be widely deployed in multiple vertical industries with high requirements for data sovereignty and system security assurance. The following are typical deployment embodiments of the present invention in different vertical industries:

1. Industrial manufacturing scenario: For intelligent equipment defect inspection in smart factories, targeting the pain points of loss of industrial design drawing data sovereignty and cold boot side-channel charge residue extraction attacks pointed out in the Background Art, in this embodiment, the **Isolated Asset Layer** is deployed in a local private Industrial Control System (DCS) of the factory to store un-desensitized inspection image features and equipment fault expert parameters; the **Rule Cognition Layer** is deployed in a **trusted isolated operating environment** of an edge industrial control security gateway in the factory area to dynamically open an **isolated secure operating domain**; the **Computing Execution Layer** is deployed on a third-party cloud GPU computing platform to run a large-scale visual defect detection model. The **Rule Cognition Layer** extracts local image feature data via a **unidirectional read-only physical tunnel**, assembles inspection inference instructions within the isolated domain, and transmits the instructions to the cloud for inference unidirectionally. Once detection is completed

or a timeout occurs, **physical degaussing erasure** is performed on the isolated domain memory of the edge gateway to prevent leakage of core industrial design drawings and defect parameters to the public cloud. Based on the three-layer separated architecture, this scenario physically eliminates exposure risks of core industrial data on cloud computing nodes and cold boot side-channel extraction risks, and the scenario is implemented by any operation method according to **Claim 1** to **15** of the present invention.

2. Government confidential scenario: For intelligent official document approval in e-government affairs, targeting the pain points of loss of national sensitive data sovereignty and abnormal boundary penetration security risks of untrusted cloud computing nodes pointed out in the Background Art, in this embodiment, the **Isolated Asset Layer** is deployed in a physical sovereign machine room of the government intranet to store top-secret official documents and basic civil affairs archives; the **Rule Cognition Layer** is deployed on a confidential computing all-in-one machine at the boundary of the party and government intranet; the **Computing Execution Layer** is deployed on a public LLM node of the national e-government extranet cloud platform. The **Rule Cognition Layer** extracts desensitized document slices and injects them into an **isolated secure operating domain** to convert them into document outline instruction streams for transmission to the public LLM for inference. Kernel-level page fault monitoring via eBPF prevents illegal reading by the cloud or bypass side channels. Once the session ends, **N_scrub**=5 rounds of physical overwriting degaussing are immediately executed to neutralize DRAM residual charges. This scenario realizes fully controlled leakage risks of government sensitive official documents during transmission and inference, and the overall system security and data sovereignty control strength fully meet Class III and above high-security compliance requirements and confidential data security review standards of the state, and the scenario is implemented by any operation method according to **Claim 1** to **15** of the present invention.

3. Financial data scenario: For anti-money laundering suspicious transaction chain auditing, targeting the pain points of loss of plaintext customer identity sovereignty and malicious LLM **Over-ingestion Attack** unauthorized attacks on cloud inference engines pointed out in the Background Art, in this embodiment, the **Isolated Asset Layer** is deployed in a local secure zone of a commercial bank core database to store customer real-name identity (KYC) and plaintext transaction flows; the **Rule Cognition Layer** is deployed in a confidential enclave of a branch front-end gateway; the **Computing Execution Layer** is deployed on a large-scale compliant inference cluster of the head office or affiliated fintech group. The **Rule Cognition Layer** aggregates sensitive money laundering slices with standardized cognitive protocol instructions to generate control graph instruction streams for inference and calculates an HMAC-SHA256 **instruction anti-replay digital fingerprint** for unidirectional transmission to the inference

cluster. If the inference cluster triggers an **Over-ingestion Attack** due to abnormal scanning (excessive Token consumption), the gateway actively fuses and sends a TCP RST packet with Window Size = **0xABCD** for evidence solidification. This scenario realizes credible judgment of suspicious money laundering chains within an extremely short time, and millisecond-level automatic fusing and physical lock-in forensics when **Over-ingestion Attacks** are detected, and the scenario is implemented by any operation method according to **Claim 1** to **15** of the present invention.

Finally, those skilled in the art can understand that the core inventive concepts of the present invention such as "**unidirectional read-only physical tunnel**", "**isolated secure operating domain**", "**trusted isolated operating environment**", "**physical degaussing erasure**" and "**instruction anti-replay digital fingerprint**" are not limited to the specific software and hardware structural forms listed in the embodiments, but cover all equivalent substitutions and logical variants foreseeable by those skilled in the art without departing from the spirit and essence of the present invention, all of which shall fall within the protection scope of the present invention.